

DATA STRUCTURES - Examination 2024

Total Marks: 70

SECTION A

1.

Attempt all questions in brief.

- (a) Define array and linked list.
- (b) What is a circular queue?
- (c) State the time complexity of merge sort.
- (d) What is a stack? Give two applications.
- (e) What is a binary search tree?
- (f) Differentiate between BFS and DFS.
- (g) What is a hash function?

SECTION B

2.

Attempt any three of the following:

- (a) Differentiate between array and linked list with suitable examples.
- (b) Explain stack data structure with push and pop algorithms.
- (c) Explain inorder, preorder and postorder traversal of a tree.
- (d) Explain hashing and different collision handling methods.
- (e) Explain merge sort and analyze its time complexity.

SECTION C

3.

Attempt any one part of the following:

- (a) Explain binary search tree with insertion and deletion examples.
- (b) Explain BFS and DFS traversal of a graph with an example.

4.

Attempt any one part of the following:

- (a) Write the binary search algorithm and discuss its complexity.
- (b) Write a short note on AVL trees and rotations.